WORK DONE

Work Done (Joules) = Distance Travelled (metres) x Force Applied (Newtons)

a) A crane lifts a 100N object 10 metres above the ground. What is the work done?

b) An athlete runs 100 metres against a frictional force of 5N. What is the work done?

c) A weightlifter lifts 100Kg 1.5 metres above the ground. What is the work done?

d) A 7500N Motorbike travels 1 kilometre. What is the work done?

e) A 12g table tennis ball is hit 2.5 metres. It is then returned. What is the total work done by the ball?

f) A 75 Kg person climbs a ladder to 7.5 metres above the ground. What is the work done?

g) A lift carries 6 people weighing 420 Kg up to 12 metres. It then picks up a further two more people both weighing 120 Kg and lifts everyone up 6 more metres. What is the total work done?